

```

$ curl https://raw.githubusercontent.com/kubernetes/helm/master/scripts/get > get_helm.sh
$ chmod 700 get_helm.sh
$ ./get_helm.sh
Total 100 6689 100 6689 0 0 5669 0 0:00:01 0:00:01 --:--:-- 5673
helm v2.8.2 is already latest
helm installed into /usr/local/bin/helm
Run 'helm init' to configure helm.
$

```

Figure 9: Installing Helm

To start Helm, execute the following command:

```
helm init
```

This will create a server as a pod and configure the cluster to run Helm. There are many applications that can be installed using Helm. An exhaustive list of Helm-supported applications can be found at Kubeapps Hub [6].

Let us install the Kubernetes dashboard by running the commands shown in Figure 10.

This will install the dashboard application as a pod on

```

$ helm install stable/kubernetes-dashboard --name osfy-simple-app
NAME: osfy-simple-app
LAST DEPLOYED: Wed Mar 28 18:59:31 2018
NAMESPACE: default
STATUS: DEPLOYED

RESOURCES:
==> v1/Service
NAME: osfy-simple-app-kubernetes-dashboard
TYPE: ClusterIP
CLUSTER-IP: 10.107.128.56
EXTERNAL-IP: <none>
PORT(S): 80/TCP
OS:

==> v1beta1/Deployment
NAME: osfy-simple-app-kubernetes-dashboard
DESIRED: 1
CURRENT: 1
UP-TO-DATE: 1
AVAILABLE: 1
AGE: 0s

==> v1/Pod(related)
NAME: osfy-simple-app-kubernetes-dashboard-cdc6c68c7-bn7k
READY: 0/1
STATUS: ContainerCreating
RESTARTS: 0
AGE: 0s

```

Figure 10: Deploying applications on Helm

the Kubernetes cluster. After a while, running the command from Figure 6, you will see a pod running with the name 'osfy-simple-app-kubernetes-dashboard-xxxxx-xx'. The last few characters are randomly assigned by the cluster itself.

Once the pod is running, execute the commands as shown in Figure 11 to access the application. These commands are needed to expose the port of the dashboard to your host, so as to access the application from your host.

After a few seconds, open the URL <http://localhost:9090> in your browser. You will find the dashboard of your Kubernetes cluster as shown in Figure 12. The dashboard has three figures on the top, which indicate the running status of

```

$ export POD_NAME=$(kubectl get pods -n default -l "app=kubernetes-dashboard,release=osfy-simple-app" --no-headers | awk '{print $1}')
$ echo http://localhost:9090/
http://localhost:9090/
$ kubectl -n default port-forward $POD_NAME 9090:9090
Forwarding from 127.0.0.1:9090 -> 9090
Forwarding from [::]:9090 -> 9090

```

Figure 11: Running the dashboard application

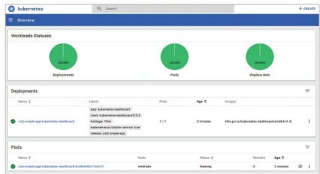


Figure 12: Kubernetes dashboard application

all the deployments, pods and replica sets in the cluster. The other things on the dashboard are self-explanatory.

Sometimes, we may want to delete the applications deployed for various reasons. To do so, run the following command:

```
helm delete osfy-simple-app
```

As you have seen, it is easy to run a Kubernetes cluster locally, and Helm makes it simple to deploy applications on the cluster. Helm charts make it easier for anyone to package their applications and deploy them on the cluster. In this article, we have only discussed deploying applications, but once that is done, in a later article we can look into the features of Kubernetes—like self-healing, replica-sets, scaling the pods, stateful sets, custom resource definitions, etc. 

References

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